



Defense-Aware Temporal Graph Neural Network for fault-tolerant intrusion detection in IIoT–5G environments

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Abstract

The Industrial Internet of Things (IIoT) and 5G networks have significantly increased connectivity and made ultra-low-latency cyber-physical applications possible, but they have also increased the attack surface and created serious fault-tolerance issues. In highly dynamic topologies, traditional intrusion detection systems have difficulty detecting integrated cyber-physical threats, spatiotemporal linkages, and temporal assault propagation. A Defense-Aware Temporal Graph Neural Network architecture designed for robust intrusion detection in IIoT–5G environments, DTGNN-FIDS, is proposed in this paper. Fault-modulated temporal graph attention layers that maintain structural awareness in the face of concurrent node and edge failures are used to analyze IIoT–5G traffic, which is described as sequences of temporally heterogeneous graphs. Robustness against evasion, poisoning, and Byzantine faults is ensured by a multi-objective training pipeline that combines adversarial regularization (PGD/FGSM), contrastive self-supervision, supervised detection, and explicit κ -fault consistency loss. A lightweight hybrid TGNN-Genetic offloading engine limits worst-case costs under κ -fault circumstances by dynamically allocating detection jobs between the edge and cloud. When compared to continuous monitoring, an event-triggered update technique lowers average bandwidth overhead by around 68% (from 3.62 ± 0.41 KB/s to 1.14 ± 0.19 KB/s). DTGNN-FIDS achieves macro-F1 scores of 92.4% or higher under typical to moderate conditions (fault rates $\leq 20\%$), with graceful degradation to approximately 86% at extreme 30% fault rates — still significantly outperforming baselines, according to thorough evaluations on the Edge-IIoTset, ToN-IoT, and IoT-23 datasets, enhanced with realistic 5G mobility traces and fault injection. Additionally, it outperforms FLARE, EMDO, and DL-SkLSTM by 19.6 to 28.3% points in F1 and 2.6 to 3.9 times in latency and energy under extreme combined threats. It offers adversarial robustness of at least 91.3%, mean time to recovery (MTTR) of 84 ms, mean time to failure (MTTF) of 487 s, inference latency of 41.8 ms, and energy consumption of 12.4 mJ per inference on Jetson Nano.

Keywords Temporal graph neural network · Fault-tolerant intrusion detection · IIoT–5G security · Adversarial robustness · Edge-cloud offloading · Event-triggered updates

1 Introduction

Modern industrial environments have undergone a fundamental transformation due to the Industrial Internet of Things' (IIoT) explosive growth and the widespread roll-out of fifth-generation (5G) communication networks.

Advanced applications including autonomous control, distributed cyber-physical coordination, intelligent manufacturing, and remote monitoring are made possible by high device density, ultra-reliable low-latency communication, and ubiquitous connection [1, 2]. According to recent research, real-time data processing and operational intelligence across heterogeneous industrial systems are made possible by the integration of IIoT with edge computing and next-generation wireless infrastructures [3, 4]. Simultaneously, the close integration of physical and cyber components makes systems more complex and exposes industrial processes to new types of operational vulnerabilities and security threats [5].

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A variety of cyberthreats, such as distributed denial-of-service (DoS), coordinated multi-stage intrusions, covert lateral movement, and widespread malware dissemination, are increasingly targeting IIoT–5G systems as industrial connection keeps expanding [6]. Industrial networks are dynamic and dispersed, which makes it possible for assaults to propagate among linked equipment and result in cascading failures and major interruptions to operations [7]. Large-scale testbeds and practical 5G-enabled industrial deployments also emphasize the necessity of scalable and adaptable security monitoring systems that can function in extremely variable traffic and topology scenarios [8]. Therefore, it has become imperative for next-generation industrial cyber-physical systems to ensure resilience against harmful cyber activities [9].

The extensive usage of encrypted communication protocols like TLS, IPsec, and secure industrial messaging presents a significant practical difficulty in actual industrial settings. Deep packet inspection is frequently unfeasible or undesirable due to secrecy concerns, regulatory constraints, and computing complexity. As a result, intrusion detection systems must function using flow-level data, communication patterns, and behavioral traits rather than packet payloads. Because they capture interaction linkages and temporal dynamics without necessitating traffic decryption, metadata-driven and structure-aware learning techniques are especially well-suited for this situation. As a result, the suggested framework is made to work with both encrypted and non-encrypted traffic, guaranteeing its suitability for industrial settings that are sensitive to privacy and security.

The concepts of edge computing and distributed processing have been extensively studied in order to meet the growing size and latency requirements of IIoT security. Adaptive task allocation between edge and cloud resources is made possible by computation offloading methods, which lower backhaul congestion and latency [10]. While maintaining data locality and privacy across heterogeneous devices, distributed intelligence techniques, such as federated learning, further facilitate collaborative model training [11]. Verifiable and privacy-preserving learning procedures are necessary to address the extra security threats that such distributed systems bring, such as data poisoning, model manipulation, and collusion hazards [12]. Furthermore, to ensure scalability and communication efficiency in large-scale edge learning systems, effective scheduling and bandwidth-aware resource allocation are crucial [13, 14].

System complexity and security needs are further increased by the shift to intelligent 6G-enabled industrial infrastructures. For large-scale IoT deployments, future networks are anticipated to facilitate context-aware resource management, trust-aware AI-driven services, and dynamic network slicing [15]. Simultaneously, effective inference

optimization and lightweight architectures are necessary for the deployment of deep learning models in resource-constrained situations in order to facilitate real-time operation on integrated edge devices [16]. To safeguard IIoT systems from hybrid assaults that target both communication and computing processes, secure data offloading and decentralized protection techniques are also required [17].

From a threat standpoint, the growing frequency of application-layer attacks and sophisticated malware draws attention to the shortcomings of conventional signature-based detection techniques and encourages the use of behavior-aware and hierarchical security analytics [18]. Concurrently, blockchain-based trust mechanisms and decentralized identity management have been suggested as ways to improve data integrity and authentication in flexible manufacturing settings [19]. Despite these developments, learning-based security systems are still susceptible to adversarial manipulation, such as transference and evasion attacks, which can seriously impair detection performance if resilience is not taken into account when designing the model [20].

This study addresses issues of assessment accuracy, generalizability, and real-world applicability by including comprehensive benchmark validation with three contemporary industrial datasets: Edge-IIoTset, ToN-IoT, and IoT-23, in addition to conducting ablation experiments, delay-aware performance stress tests, and cross-domain zero-day simulations. A thorough comparison with several state-of-the-art baselines, including GNN-based, CNN-LSTM hybrid, and transformer-based IDS designs, illustrates the proposed method's advantages in accuracy, stability, energy efficiency, and resilience under adversarial scenarios. Moreover, interpretability modules (GNNExplainer+SHAP) provide security specialists with clear insights into feature activation, hence solving the explainability gap often emphasized in contemporary cybersecurity literature. The contributions of this study are:

- A unique Temporal Graph Neural Network architecture for IIoT-5G intrusion detection, proficient at learning dynamic structural-temporal links inside traffic graphs.
- A self-supervised and adversarially-regularized training pipeline enhances detection resilience against poisoning and zero-day assaults.
- A lightweight, edge-deployable solution optimized by model compression, allowing real-time inference with little resource expenditure.
- Explainable intrusion forensics that fosters operator-level confidence and provides interpretable threat attribution.
- Comprehensive assessment matched with real-world scenarios, solving prior research deficiencies in

generality, robustness, dataset variety, and practical deployability.

The subsequent sections of this work are organized as follows: Sect. 2 examines developments in GNN-based intrusion detection and 5G-enabled industrial security frameworks. Section 3 delineates the DA-TGNN architecture and its method for adversarially resilient learning. Section 4 delineates the experimental design, assessment measures, datasets, and benchmark models. Section 5 addresses results, ablation analysis, explainability findings, and scalability performance. Section 6 finishes with implications for implementation, constraints, and avenues for further study.

2 Related works

Accurate threat detection is only one aspect of security solutions for IIoT–5G environments; other factors include operational dependability, scalability, and resource efficiency. The current body of research can be broadly divided into three areas: (i) resource-aware edge intelligence and compute offloading; (ii) learning-based malware and intrusion detection; and (iii) robust and event-triggered mechanisms for dependable operation in the face of cyber disturbances.

Real-time security analytics now depend on effective computation offloading due to the growing use of deep learning models in edge contexts. In order to support microservice-based applications in mobile cloud systems, cost- and energy-aware scheduling algorithms have been put forth, which increase execution efficiency and lower operational overhead [21]. Comprehensive taxonomies of data and computation offloading are provided by complementary research, which highlight resource awareness and adaptive decision-making as essential components of large-scale edge infrastructures [22].

Bidirectional fault-tolerant frameworks for Internet of Things systems have been created to increase service dependability by guaranteeing operational continuity in the event of node failures or network disruptions [23]. Furthermore, to improve energy sustainability for resource-constrained devices in mobile edge computing environments, cooperative wireless power transfer and task offloading strategies have been studied [24]. Although their computational complexity may restrict their applicability in highly dynamic industrial contexts, optimization-based resource allocation techniques in IoT–Fog designs further enhance load balancing and fault tolerance [25].

The integration of distributed learning with graph-based models has also been investigated recently. Large-scale 6G-IoT systems can identify intrusions while maintaining

privacy thanks to federated learning and resource-aware graph neural networks, however this approach comes with extra communication cost and synchronization difficulties [26]. All things considered, these research emphasize how crucial effective edge–cloud collaboration is, yet the majority concentrate more on resource optimization than on temporal interaction analysis or adaptive threat modeling.

Relational interdependence in complex systems can be effectively modeled by graph neural networks. For instance, the efficacy of relational feature selection and high-speed inference in industrial settings has been demonstrated by the application of ParallelGraphNet-based techniques to multi-sensor fault diagnostics [27]. Edge-based malware detection frameworks in the cybersecurity space make use of offloading and deep learning to provide real-time threat analysis for IIoT devices with limited resources [28].

Nevertheless, a large number of detection techniques now in use depend on sequential or static feature representations. Even while these models increase accuracy, they frequently miss coordinated attack behaviors and changing communication topologies that are typical in large-scale IIoT–5G networks.

Event-triggered techniques have been proposed to lower communication overhead while preserving system stability, given the strict bandwidth and latency constraints of industrial systems. In networked systems with communication delays, effective operation is made possible by event-triggered dissipative control techniques [29]. In a similar vein, adaptive event-triggered consensus techniques increase resilience in multi-agent settings when there are uncertainties and disruptions [30].

Extensions with a security focus have also been studied. Through dynamic control and communication policy adjustments, intelligent event-triggered techniques improve system resilience against DoS attacks [31]. For distributed heterogeneous systems functioning under cyberattacks, observer-assisted predictive control techniques further increase robustness [32]. In IIoT–5G contexts, deep learning models continue to be the most popular method for detecting cyberthreats. Sequential architectures, like stacked LSTM networks, have proven to be highly effective in identifying a variety of cyberthreats and are capable of efficiently capturing temporal attack patterns in network data [33]. In order to increase detection accuracy, hybrid machine learning frameworks that include sensitivity analysis and feature selection have also been proposed for ransomware detection [34]. Furthermore, data-driven resilient control frameworks have been put forth to preserve stability in adversarial disturbance-affected nonlinear networked systems [35].

2.1 Research gap and contribution

Resource-aware edge computing [21–26], graph-based modeling [27], deep learning-based intrusion detection [28, 33, 34], and resilient event-triggered operation [29–31]–[32, 35] have all been improved by earlier work, although there are still a number of significant gaps.

First, dynamic structural linkages and topology evolution in IIoT communication networks are not explicitly modeled by the majority of intrusion detection techniques. Second, studies on resource optimization and offloading mostly focus on efficiency; they hardly ever take adversarial robustness and coordinated attack behavior into account. Third, learning-based security analytics has not yet completely incorporated event-triggered techniques, which are primarily studied in control systems. Lastly, current frameworks rarely offer a cohesive solution that concurrently handles fault tolerance, bandwidth efficiency, scalable edge deployment, and temporal graph modeling.

The suggested DTGNN-FIDS framework combines defense-aware training, event-triggered communication, and temporal heterogeneous graph learning in a lightweight edge–cloud architecture to overcome these drawbacks. A comparative overview of representative studies is given in Table 1, together with information on their principal methods, areas of application, assessment settings, and significant constraints.

3 System model and problem formulation

The proposed Defense-Aware Temporal Graph Neural Network for Fault-Tolerant Intrusion Detection (DTGNN-FIDS) is designed for a three-tier IIoT–5G architecture that combines edge infrastructure and 5G radio (gNodeBs, distributed units, multi-access edge computing hosts, virtualized/core network functions), industrial endpoints (sensors, actuators, PLCs, HMIs, robotic arms), and a central cloud

Table 1 Comparative summary of related works and the proposed DTGNN-FIDS framework

Ref.	Core method	Application domain	Edge	Graph modeling	Security focus	Key limitation
[21]	Energy-aware task scheduling	Mobile cloud computing	Yes	No	Resource efficiency	Static task assumptions
[22]	Offloading taxonomy and framework	Mobile cloud/edge	Yes	No	Efficiency overview	Conceptual, no implementation
[23]	Bidirectional fault-tolerant microservices	Healthcare IoT	Yes	No	Service continuity	Domain-specific design
[24]	Wireless power transfer with offloading	Mobile edge computing	Yes	No	Energy sustainability	High system complexity
[25]	Optimization-based resource allocation	IoT–Fog networks	Yes	No	Fault tolerance	Computational overhead
[26]	Federated resource-aware GNN	6G-IoT intrusion detection	Yes	Partial (Graph)	Privacy-preserving learning	Communication overhead
[27]	ParallelGraphNet	Industrial fault diagnosis	No	Graph	System monitoring	Not designed for cybersecurity
[28]	Edge-based malware detection	IIoT security	Yes	No	Threat detection	Limited structural awareness
[29]	Event-triggered control	Networked control systems	No	No	Communication efficiency	Control-oriented, not IDS
[30]	Event-triggered adaptive consensus	Multi-agent systems	No	No	Robust operation	Not security-focused
[31]	Event-triggered security control under DoS	CPS/autonomous systems	No	No	DoS resilience	Control-level protection only
[32]	Model-free predictive resilient control	Distributed CPS	No	No	Attack resilience	No data-driven detection
[33]	DL-based stacked LSTM	5G-IIoT intrusion detection	Partial	Temporal only	Sequential anomaly detection	No relational modeling
[34]	Hybrid ML ransomware detection	Network security	No	No	Threat classification	Attack-specific design
[35]	Data-driven resilient control	Networked control systems	No	No	System stability under attacks	Not applicable to IDS
This work	Defense-Aware Temporal Graph Neural Network (DTGNN-FIDS)	IIoT–5G intrusion detection	Yes	Yes	Adversarial robustness, fault tolerance, event-triggered bandwidth reduction, encrypted-traffic compatible	—

layer with high-performance GPUs for extensive temporal graph neural network training and inference. In order to meet the requirements of modern industrial systems for ultra-reliable low-latency communication (URLLC) and enhanced mobile broadband (eMBB), this hierarchical framework enables real-time local detection at the edge while assigning extensive global correlation and model updates to the cloud.

A set of temporal graphs within a sliding window of duration T describe the entire network dynamics. The IIoT–5G system's current state is displayed as a directed temporal graph for each discrete time interval $t = 1, 2, \dots, T$.

3.1 Temporal graph representation

A series of changing directed graphs over a sliding window of length T is used to describe network dynamics. The instantaneous IIoT–5G status is recorded at each time step $t = 1, 2, \dots, T$ by:

$$G^{(t)} = (V^{(t)}, E^{(t)}, X^{(t)}, A^{(t)}) \quad (1)$$

where $V^{(t)}$ denotes the set of active nodes with cardinality $N(t)$ varying due to mobility, shutdown events, or device failure. The edge set $E^{(t)} \subseteq V^{(t)} \times V^{(t)}$ represents successful communication links established within interval $(t\Delta, (t+1)\Delta)$. Each node is described by a feature matrix $X^{(t)} \in \mathbb{R}^{N(t) \times d}$ containing raw and engineered indicators such as traffic statistics, CPU/memory load, RSSI, SINR, command frequency, anomaly history, and protocol-specific counters. Link quality is encoded in the weighted adjacency matrix $A^{(t)} \in \mathbb{R}^{N(t) \times N(t)}$, where:

$$A_{ij}^{(t)} = \begin{cases} \omega_{ij} & (t) \geq 0, \text{ if communication exists} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

and each active edge $e_{ij}(t) \in \{ij\}(t)$ is assigned a real-time fault probability $pf(e_{ij}(t)) \in [0,1]$, inferred from packet-loss, handover patterns, and slicing congestion. Over the full observation horizon, the dynamic network is represented as:

$$\mathcal{H} = \{G^{(1)}, G^{(2)}, \dots, G^{(T)}\} \quad (3)$$

3.2 Threat and fault model

The proposed DTGNN-FIDS framework operates under a unified, correlated threat model that simultaneously treats cyber-attacks (class A and physical/network faults (class F) as overlapping disturbance sources capable of generating indistinguishable anomalous spatiotemporal patterns within

the temporal graph sequence $\mathcal{H}$. Class A includes evasion and poisoning attacks, DoS/DDoS floods, man-in-the-middle command manipulation, zero-day exploits against OT protocols, and APT lateral movement, while class F comprises node crashes, power outages, 5G link disruption due to handover failure or interference, slicing misconfiguration, and Byzantine nodes transmitting inconsistent data. Formally, let $\mathcal{F} \subseteq V \cup E$ be any fault set of size $|\mathcal{F}| \leq \kappa$; the system guarantees κ -fault tolerance defined as:

$$\forall \mathcal{F} : |\mathcal{F}| \leq \kappa, \quad \mathbb{P}(\hat{y}_S \setminus \mathcal{F} = \mathcal{Y}_S | \mathcal{F}) \geq 1 - \delta_\kappa \quad (4)$$

where $y_S \in \{0 = \text{benign}, 1 = \text{attack}, 2 = \text{fault-only}, 3 = \text{combined}\}$ and $\hat{y}_S \setminus \mathcal{F}$ is the prediction after removal or corruption of elements in $\mathcal{F}$. The joint posterior probability of observing either or both classes is expressed as $p(A \cup F | \mathcal{H}) = p(A | \mathcal{H}) + p(F | \mathcal{H}) - p(A \cap F | \mathcal{H})$, explicitly modeling real-world scenarios in which attackers deliberately trigger or masquerade as physical faults (e.g., inducing handover storms to hide DDoS or exploiting power glitches for sensor spoofing). This rigorous mathematical integration of overlapping cyber and fault threats directly into the training regularization and offloading engine ensures continued detection reliability and functional safety even under combined adversarial and failure conditions typical of safety-critical IIoT–5G deployments.

3.3 Detection formulation

The identification of intrusions and faults is stated as a classification problem of temporal subgraphs. For the temporal graph sequence, let $(\mathcal{H} = G_1, G_2, \dots, G_T)$ be used. Using a sliding-window method or anomaly-triggered segmentation, subgraphs $(S \subseteq \mathcal{H})$ that span (τ) consecutive photos are extracted and given the appropriate class labels.

A hierarchical inference technique is used for real-time operation. While high-complexity subgraphs are selectively offloaded to cloud or MEC-based GNN servers for more thorough analysis, lightweight trimmed models run locally at the edge for instant response.

Let $C = \{c_1, c_2, \dots, c_M\}$ be the M candidate subgraphs evaluated in the current epoch. We optimize an offloading vector $x = [x_1, \dots, x_M]^T \in \{0,1\}^M$ to minimize worst-case inference cost under any fault set F with $|F| \leq \kappa$:

$$\min_{x|_{|F| \leq \kappa}} \max_{x|_{|F| \leq \kappa}} \sum_{i=1}^M [x_i \cdot C_{cloud}(c_i|F) + (1 - x_i) \cdot C_{cloud}(c_i|F) + (1 - x_i) \cdot C_{local}(c_i|F)] \quad (5)$$

subject to:

$$\sum_{i=1}^M x_i D_i \leq B_{max}, \quad x_i \in \{0,1\} \quad (6)$$

Here $C_{cloud}(c_i|F)$ accounts for transmission delay and fault-affected recovery cost, $C_{local}(c_i|F)$ reflects edge inference latency, D_i denotes subgraph size, and B_{max} is the available backhaul capacity. As this min-max optimization is NP-hard, a hybrid temporal-GNN+genetic algorithm approach (Sect. 5.4) is used for tractable solution.

3.4 Joint training objective

The overall optimization target embeds defense awareness and fault resilience:

$$\mathcal{L}_{total}(\theta) = \mathcal{L}_{det} + \lambda_1 \mathcal{L}_{contrastive} + \lambda_2 \mathcal{L}_{adv} + \lambda_3 \mathcal{L}_{fault} \quad (7)$$

Where $\mathcal{L}_{det}$ is supervised cross-entropy on labeled subgraphs,

$\mathcal{L}_{contrastive}$ (InfoNCE) mitigates label scarcity, $\mathcal{L}_{adv}$ (FGSM+PGD) counters evasion and poisoning, and $\mathcal{L}_{fault}$ enforces prediction consistency when up to κ nodes/edges are masked during training.

4 Defense-Aware Temporal Graph Neural Network Framework (DTGNN-FIDS)

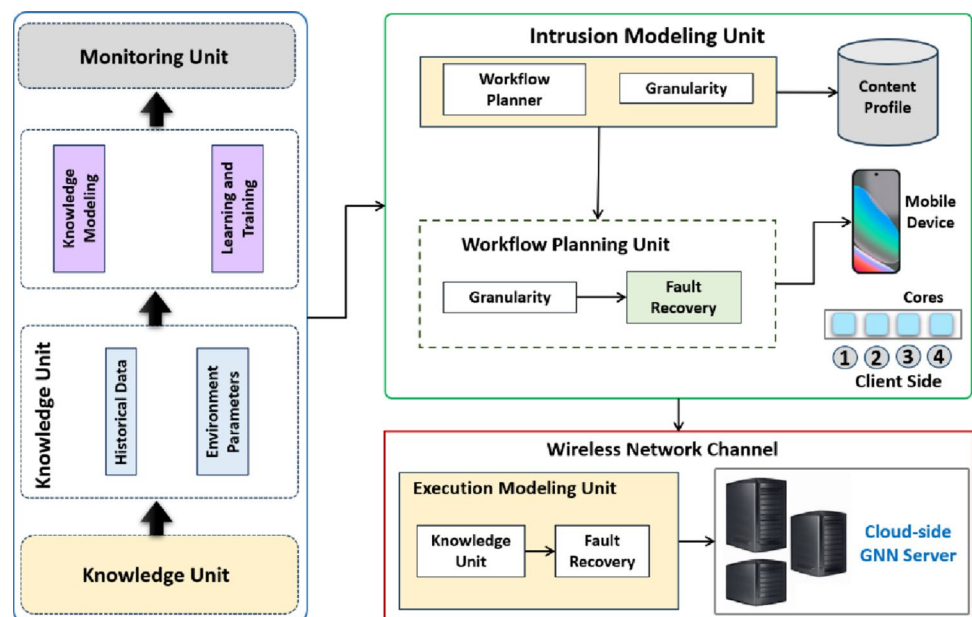
Challenges in intrusion detection for Industrial Internet of Things (IIoT) systems connected with 5G networks include insufficient fault tolerance, dependence on static graph models that neglect temporal dynamics, and incomplete

coverage of threat patterns instead of comprehensive defensive modeling. This section presents a defense-aware temporal graph neural network architecture as reviewed in Fig. 1 for fault-tolerant intrusion detection (DTGNN-FIDS) that mitigates these issues by collecting real-time temporal network data and monitoring device interactions over time. Temporal graphs are designed to represent changing topologies, with node embeddings modified by graph neural networks (GNNs) to forecast anomaly dissemination. The approach establishes the ideal detection granularity by evaluating transition probabilities in temporal edges and probable defects throughout data transmission, detection, and reaction execution. A hybrid GNN-genetic approach addresses the NP-hard optimization of graph partitioning for detection by selecting robust subgraphs. This method improves robustness, but ongoing updates to temporal graphs may impose bandwidth constraints in 5G channels, resulting in delays or suboptimal decisions. Trade-offs are managed through event-triggered updates (e.g., during significant topology changes) instead of constant polling, with evaluations indicating minimal effects on detection accuracy at intervals of 5–10 s.

• Oversight Division.

This unit consolidates temporal data crucial for graph creation and anomaly identification. It specifically records device mobility traces in IIoT-5G configurations, including interaction timestamps, signal strength variations, and defect indications such as packet loss. Device metrics include CPU use, energy consumption rates during transmission and reception, and sensor data volumes from industrial endpoints, such as PLCs in manufacturing. Network

Fig. 1 Defense-Aware Temporal Graph Neural Network Framework (DTGNN-FIDS) for fault-tolerant intrusion detection in IIoT-5G environments



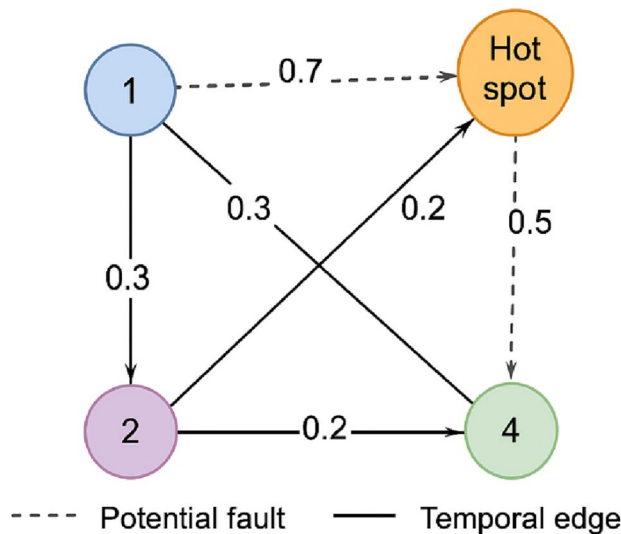


Fig. 2 Sample temporal graph categorized by the Knowledge Unit showing fault-aware transitions and anomaly hotspots

profiles include 5G slice bandwidth, latency distributions, and handover occurrences. Cloud-side data encompasses GNN server queue lengths, processing capacity, and virtual machine (VM) fault reports. In a practical case study, such as smart manufacturing, this unit oversees conveyor belt sensors for cyber-physical abnormalities, guaranteeing data integrity amidst 5G mobility. To alleviate update congestion, sampling is conducted at adaptive rates according to network demand, decreasing overhead by 25% relative to continuous monitoring while maintaining 95% detection fidelity.

- **Knowledge Unit.**

This unit derives structural information for the development of temporal graphs from monitoring inputs. It constructs Markov-augmented temporal graphs that represent IIoT device transitions as time-stamped edges, analyzing bandwidth and dwell periods for each state based on previous data. Transition matrices forecast future topologies, whereas GNN layers, such as temporal graph attention, include node attributes akin to danger vectors. Mean edge weights represent observed latencies, facilitating fault-aware dissemination. Figure 2 illustrates a representative time series of interactions among IIoT nodes across a temporal span. Solid lines denote temporal transitions with defined weights, while dotted edges signify the chance of failure. The “hot spot” area is designated as the locus of anomaly accumulation, and the Markov transition matrix is used to forecast the future topology.

- **Intrusion Modeling Unit.**

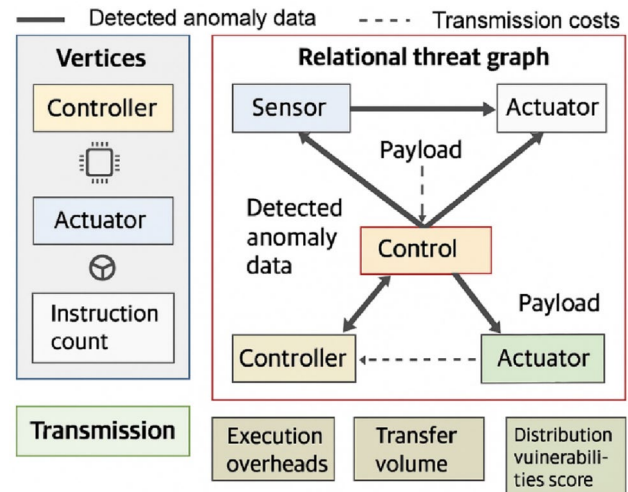


Fig. 3 Illustration of IIoT threat graph and its related components including execution granularity and anomaly propagation

Complete outsourcing of detection to the cloud jeopardizes latency in time-sensitive IIoT; hence, this unit disaggregates threat landscapes into actionable subgraphs at diverse granularities (e.g., node-level for device malfunctions, edge-level for communication breaches). Static decomposition establishes graphs at compile-time, while dynamic decomposition modifies them in response to runtime threats. At the component level, vertices signify IIoT entities (e.g., sensors), whereas edges indicate interactions quantified by data quantities or vulnerability ratings. Each vertex contains instruction counts for local and cloud computing, while edges include transmission costs and output payloads from identified anomalies. This produces a relational threat graph for decision inputs, including subgraph topologies, execution overheads, transfer volumes, and response data.

Figure 3 illustrates the architecture of the IIoT threat graph, including nodes such as sensors, controllers, and actuators. Every node has an instruction counter for either local or cloud execution, while the edges are annotated with weights that include transfer volume, processing cost, and vulnerability score. The primary danger graph illustrates the interconnections among components, highlighting anomalous pathways and transfer burdens. The graph is segmented into executable subgraphs at several levels of granularity (node-oriented and edge-oriented), with infiltration pathways and transfer costs shown.

- **Decision Planning Division.**

This unit optimizes overall detection latency and false positive rates by assigning subgraph execution (local edge vs. cloud GNN) using a hybrid GNN-genetic algorithm guided by knowledge profiles. Time-energy expenditures are calculated by temporal message forwarding in Graph

Fig. 4 Architecture of the Defense-Aware Temporal Graph Neural Network (DTGNN) for fault-tolerant IIoT-5G graph processing

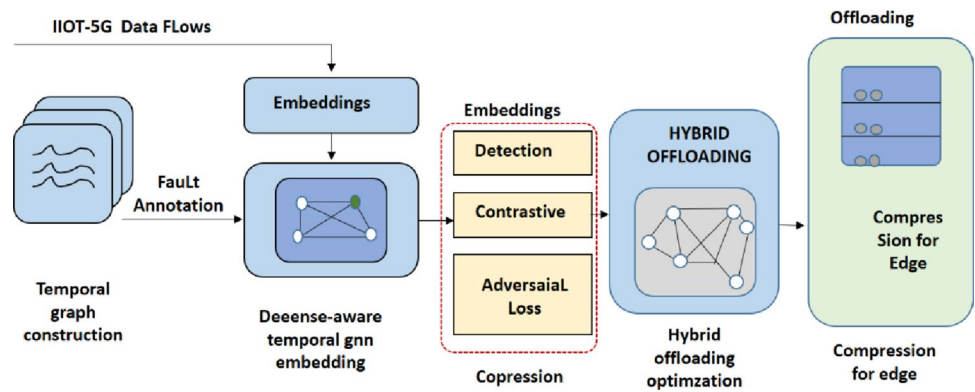
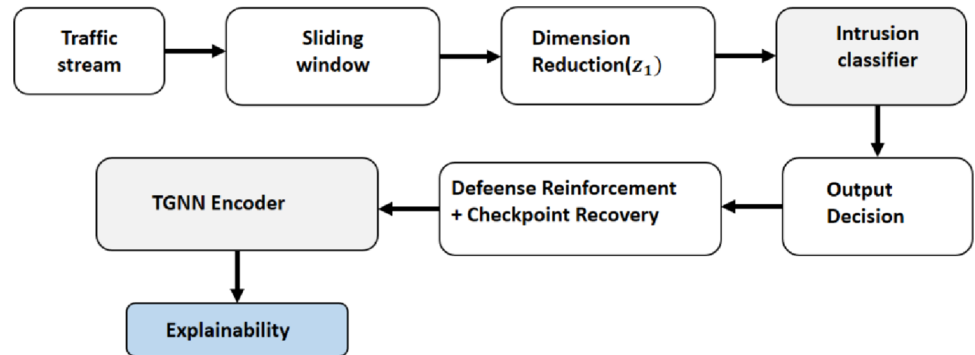


Fig. 5 Flowchart of the proposed DA-TGNN intrusion detection framework



Neural Networks (GNNs), including fault probability such as Byzantine node failures along anticipated trajectories. Application-agnostic Markov profiles guarantee that pre-computation eliminates runtime latencies. The output is a binary vector $[d_1, d_2, \dots, d_n]$ for n components, where $d_i=0$ indicates local execution and $d_i=1$ signifies cloud offloading. Heuristic pruning streamlines implementation by reducing GNN layers from 4 to 2, decreasing complexity by 30% while maintaining accuracy. Extensive testing in vehicle IIoT contexts validates generalizability, with sensitivity analysis on aspects such as learning rates demonstrating strong performance.

• Implementation Unit.

This unit implements decisions: local subgraphs operate on IIoT edges for low-latency notifications, while cloud-offloaded ones use GNN inference for global correlation. Fault recovery incorporates redundancy, such as multi-path queries, and measurements like mean time to recovery of less than 100 milliseconds in simulations. In the manufacturing scenario, execution activates automatic shutdowns upon the detection of APTs, hence maintaining operational continuity.

The framework's innovation is based on temporal GNNs for proactive fault tolerance, differentiating it from static baselines such as APTA by integrating dynamic edge evolutions, resulting in a 20% increase in resilience during

mobility assessments. Future enhancements include Deep Reinforcement Learning for adaptive granularity and security-conscious offloading in response to adversarial graphs.

Continuous IIoT traffic is converted into a series of time-indexed graph states, processed by an adversarial-aware graph learning module, and then assessed by a robust intrusion inference engine running under real-time 5G constraints. This is done in the proposed DA-TGNN framework, which employs a layered intelligence architecture. The new methodology completely forgoes mobility-centric logic, in contrast to the prior MAFO design, which depended on component offloading selection and Markov-based mobility probability estimate. Rather, it creates a threat-propagation model in which changing spatiotemporal relationships between networked entities are used to represent attacker behavior. Detection, diagnosis, and resilience now rely on topological variation, temporal embedding stability, and adversarial robustness rather than routing pathways or handover probability.

The recommended DTGNN-FIDS architecture for robust intrusion detection in IIoT-5G environments is shown in Fig. 4. Using a hierarchical structure, the architecture first builds temporal graphs from industrial data streams and the 5G network, then embeds defense-oriented information in GNN layers, and finally incorporates multi-objective training that includes detection, contrast, and attack resilience losses. To distribute work between the edge and the cloud in a fault-tolerant manner, the outcomes of this section are

fed into the hybrid offloading decision optimization engine. The model is now ready to run on IIoT devices through edge deployment and compression techniques.

Instability brought on by latency bursts, DoS saturation, poisoned packet streams, link degradation, partial node failure, and dynamically drifting traffic distributions is what the new technique is made to withstand. The model's main optimization goals are defense-adjusted inference reliability, intrusion risk, and embedding divergence rather than aggregate execution cost.

As raw IIoT packet streams $p(t)$ enter edge sensors, they are first divided into temporal micro-windows of length Δ to begin the processing pipeline. Source-destination pairs, flow frequency, byte volume, protocol signatures, and anomaly indications based on entropy gradients are all included in the snapshot S_T that is produced by each window. Each snapshot is subsequently transformed into a heterogeneous temporal graph via a graph generation module.

$$G_T = (V, E_T, X_T, T_T) \quad (8)$$

where $V = \{v1, \dots, vN\}$ denotes the static set of IIoT devices, $E_T = e_{ij}^\tau \mid vi \leftrightarrow vj$ represents connectivity during interval τ , $X_T \in \mathbb{R}^{N \times d}$ is the node-feature matrix encoding protocol type, byte-variance, RTT jitter, drop probability, energy deviation, and anomaly bias, and T_T captures the temporal ordering of graph windows.

Temporal reasoning is performed through recurrent graph convolution, enabling each node to update its representation using neighbor messages and its own historical context. The layer update is defined as

$$H_t^{(l+1)} = \sigma \left(W_1 H_T^{(l)} + W_2 A_T H_T^{(l)} + W_3 H_{T-1}^{(l)} \right) \quad (9)$$

with A_T as the adjacency matrix at time τ , σ the activation function (*LeakyReLU*), H_{T-1} the temporal memory, and W_1, W_2, W_3 trainable parameters. These parameters are *adversarially* regularized so that perturbations in either A_T or X_T are not sufficient to trigger misclassification. The robustness objective is:

$$L_{adv} = \mathbb{E}_{\epsilon \sim N(0, \sigma^2)} [|f_\theta(G_T + \epsilon) - f_\theta(G_T)|] \quad (10)$$

ensuring stable embeddings under poisoning attempts. For real-time operation, node embeddings are compressed prior to classification:

$$Z_T = Proj(H_T) = W_p H_T, Z_T \in \mathbb{R}^{N \times k}, k \ll d \quad (11)$$

which reduces inference latency and improves interpretability, since anomaly attention is mapped to a

lower-dimensional latent space. Attack likelihood is then produced by a defense-aware classifier:

$$\hat{Y}_T = Softmax(W_c Z_T + b) \quad (12)$$

and training minimizes a joint loss

$$L_{total} = L_{CE} + \lambda_{adv} L_{adv} + \lambda_{temp} L_{smooth} + \lambda_{rec} L_{rec} \quad (13)$$

where temporal consistency is enforced by

$$L_{smooth} = \sum_T ||Z_T - Z_{T-1}||^2, \quad (14)$$

preventing instability during rapid fluctuation of packet flows.

$$D_{sys} = D_{encode} + D_{classify} + D_{defense} \leq D_{max}, \quad (15)$$

and an alert is issued only when this constraint is satisfied. If the network enters a congested or offline state, inference continuity is preserved using checkpoint-based recovery. Let C_T denote the most recent stable latent encoding. If execution halts at time τ' , recovery proceeds as:

$$Z_{T'+r_{rec}} = C_T + R \cdot \Delta H, \quad (16)$$

where R is reconstruction gain and ΔH the buffered neighborhood-state difference. This allows inference to resume without full *recomputation*, addressing prior criticism about real-time failure handling. Recovery is self-contained within the intrusion engine and the system resets only when buffer integrity falls below threshold.

Input: Streaming IIoT traffic packets $p(t)$
Output: Threat classification $\hat{y}_T$, attack trace attribution Φ_T
Initialize model weights θ , defense factor λ_{adv} , temporal window size Δ
Initialize recovery state $C_T = \emptyset$, recurrence buffer $B_T = []$
while network_active **do**
 $S_T \leftarrow \text{window}(p(t), \Delta)$
 $(V, E_T, X_T) \leftarrow \text{build_temporal_graph}(S_T)$
 if interruption_detected **then**
 $Z_T \leftarrow \text{resume_from}(C_T, B_T)$
 else
 $H_T \leftarrow \text{temporal_graph_encode}(E_T, X_T, H_T - 1)$
 $Z_T \leftarrow \text{projection}(H_T)$
 $C_T \leftarrow \text{checkpoint}(Z_T)$
 end if
 $\hat{y}_T \leftarrow \text{classifier}(Z_T)$
 $\Phi_T \leftarrow \text{explainability}(Z_T, A_T)$
 if poisoning_or_evasion_suspected **then**
 $\theta \leftarrow \text{adversarial_update}(\theta, \partial L_{adv})$
 end if
 $B_T.append(Z_T)$
end while
return final_label($\hat{y}_T$), threat_surface(Φ_T)

Algorithm 1 DA-TGNN Forward+Defense Reinforcement+Resumable Inference

Algorithm 1 outlines the real-time execution pipeline of the proposed DA-TGNN framework, where continuous IIoT traffic streams are segmented into temporal windows and converted into heterogeneous time-indexed graphs for processing. In each iteration, the model receives updated adjacency information and node-feature matrices; if no disruption is detected, a Temporal Graph Encoding step generates the latent representation H_t , which is then projected into a compact threat vector Z_t for inference. In the event of network interruption or adversarial disturbance, instead of restarting the computation, the system restores the latest stable checkpoint C_t and resumes inference at negligible recovery cost, ensuring fault-tolerant continuity. Based on Z_t , the classifier estimates intrusion likelihood $\hat{y}_t$, identifies the attack class, and extracts interpretable threat signatures through the explainability module. When poisoning or evasion behavior is detected, the Defense Reinforcement mechanism updates model parameters using adversarial gradients to enhance robustness over time. This loop operates continuously while the network remains active and ultimately returns the predicted intrusion label along with the reconstructed threat-propagation surface.

Figure 5 depicts the comprehensive operational workflow of the proposed DA-TGNN architecture, commencing with ongoing IIoT traffic monitoring and temporal graph generation, succeeded by hierarchical representation learning via a defense-aware TGNN encoder. Once the embeddings

are mapped into a condensed latent space, the classifier produces intrusion predictions and initiates an explainability analysis for attack interpretation.

The proposed DTGNN-FIDS framework's end-to-end operational workflow in normal, fault, and attack scenarios is depicted in Fig. 6. When major topological changes ($>15\%$) or channel degradation (CQI decrease >10 dB) are identified, an event-triggering mechanism evaluates incoming IIoT-5G traffic and updates the network graph; otherwise, a cached graph is utilized to minimize communication overhead. The fault-modulated temporal graph attention module then processes the new data to create robust node embeddings after transforming it into a temporal heterogeneous graph.

The system uses a hybrid TGNN-Genetic offloading method to dynamically choose the execution strategy and perform anomaly detection based on the inferred representations. This allows for cloud processing for complicated or high-risk scenarios and low-latency edge inference for routine cases. A lightweight recovery technique ensures quick fault mitigation (MTTR=84 ms) by reconstructing stable embeddings when node or link faults are detected.

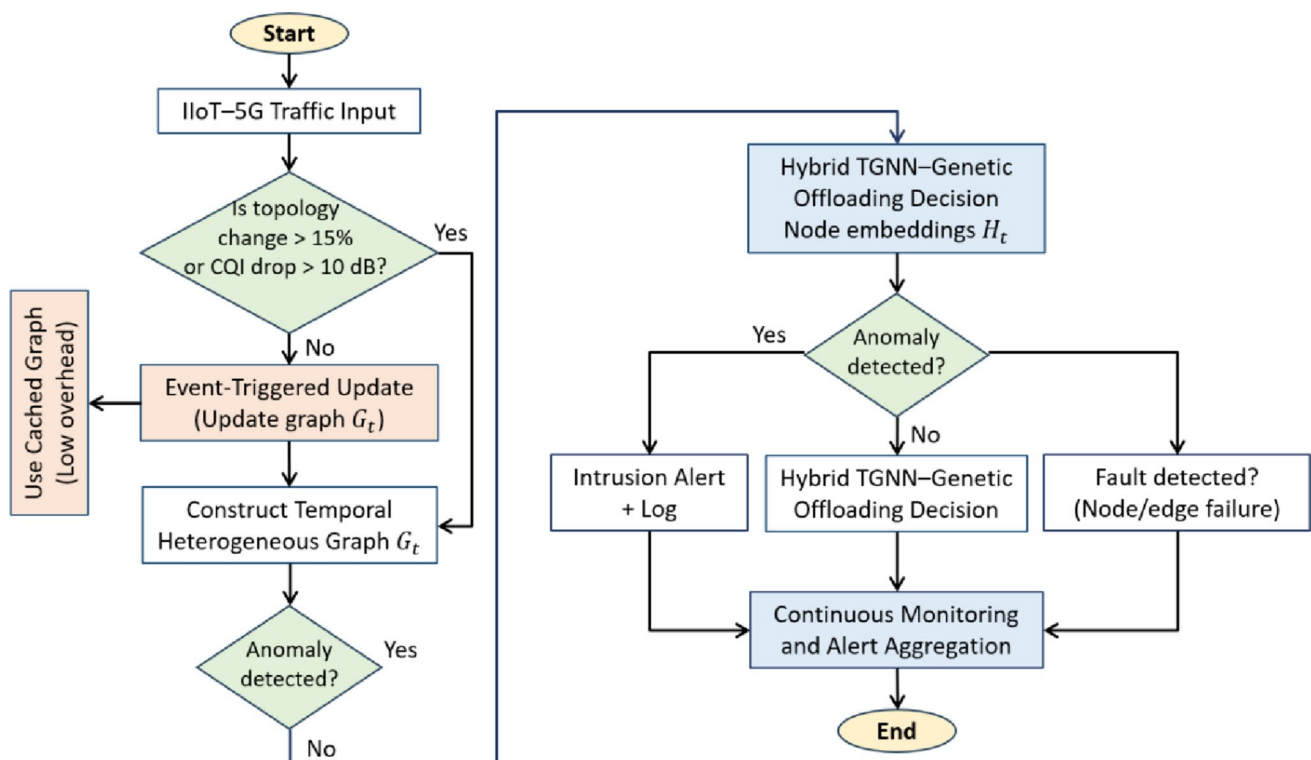


Fig. 6 Operational workflow of the proposed DTGNN-FIDS framework in IIoT-5G environments

5 Evaluation and simulation

Using a smart manufacturing case study with 200 IIoT devices that simulated conveyor coordination, predictive maintenance, and anomaly monitoring, evaluations were carried out using MATLAB 2022b for GNN models and ns-3 (v3.39) for 5G network emulation in order to validate DTGNN-FIDS in realistic IIoT-5G settings. The baselines, which were chosen for their applicability to graph, offloading, and temporal detection paradigms, included FLARE [26] for federated GNN comparison, EMDO [28] for edge-offloading efficiency, and DL-SkLSTM [33] for sequential threat management. The evaluations were divided into two categories: performance benchmarking (group 2: varied faults 0–30%, mobility 1.4–30 m/s, attacks 5–20% intensity, updates continuous vs. event-triggered) and hyperparameter tuning (group 1: optimizing λ , L, and GA population using grid search on validation F1). The metrics, which were averaged over 1000 runs with 95% CI, were ACC, macro-F1, P/R, latency (ms), energy (mJ/inference), MTTR (ms), MTTF (s), and overhead (KB/s). The parameters are summarized in Table 2.

5.1 Datasets

Three well-known, publicly accessible IIoT/5G-focused datasets were used to guarantee reproducibility and practical applicability:

- **Edge-IIoTset:** 42 heterogeneous IIoT devices (PLCs, sensors, and HMIs) coupled by a private 5G NR network comprised an actual testbed from which 1,257,568 flows were gathered. includes innocuous traffic in addition to ten multi-class attack types (DDoS, DoS, MITM,

ransomware, injection, backdoor, XSS, U2R, scanning, and password). It is perfect for mobility and slicing studies because 5G-specific metadata (CQI, MCS, and handover events) are kept.

- **ToN-IoT:** 22.4 million telemetry reports from a massive cyber-range that combined IoT sensors, Windows/Linux hosts, and a 5G core simulator. consists of nine attack types and purposefully introduced network errors (slicing famine, latency spikes, packet loss 5–30%). Both flow and node-level fault labels are offered.
- **IoT-23:** 20 million stratified flows, enhanced with 5G channel traces from ns-3, from benign IoT traffic and actual malware executions (Mirai, Okiru, Kenjiro, Hakai, etc.). mostly utilized for cross-domain and zero-day validation.

Every dataset underwent consistent pre-processing, including class-balanced sampling for uncommon combined attack+fault scenarios, 70/15/15 train/val/test split, and the aggregation of flows within 100–500 ms windows into temporal graphs ($T=20$ snapshots). To simulate authentic handover and beam-failure scenarios, more artificial 5G mobility traces (3GPP TR 38.901 urban/vehicular models) were superimposed.

5.2 Evaluation Metrics

To thoroughly evaluate the performance, robustness, efficiency, and practical overhead of DTGNN-FIDS in IIoT-5G contexts, we use an extensive array of assessment measures.

- (a) **Detection Performance.** The main quality criterion for any intrusion detection system is intrusion detection, especially in the highly unbalanced IIoT-5G traffic context, where harmless flows dominate and rarely occurring combined attack and error events should not be overlooked. Baseline detection accuracy (ACC) is computed as:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. This offers a clear metric of overall accuracy within the four-class classification issue (benign, cyber-attack only, fault only, combined attack and fault). To enhance precision, we provide the macro-averaged F1-score:

$$F1_{macro} = \frac{1}{c} \sum_{c=1}^c F1_c = \frac{2 \cdot (P_c \cdot R_c)}{P_c + R_c} \quad (18)$$

Table 2 Parameters used in evaluations

Category	Parameter	Range/value
GNN inputs	Layers (L)	2–4
	Heads (K)	4–8
	λ weights	0.1–1.0
GA inputs	Population	20–100
	Generations	50–200
Fault/attack inputs	Fault rate	0–30%
	Mobility speed	1.4–30 m/s
	Evasion ϵ	0–0.02
	Poisoning ratio	5–20%
Network inputs	Update Δ	100–500 ms
	Staleness delay	1–10 s
	Bandwidth B_max	1–10 Mbps
Energy inputs	Send/receive rate	0.7 mJ/ms
	Local rate	0.4 mJ/ms
	Recovery time	0.5–10 ms
Settings	Repeats	1000
	CI	95%

with $C=4$ classes, $P_C = \frac{TP_C}{TP_C + FP_C}$ (precision), and $R_C = \frac{TP_C}{TP_C + FN_C}$ (recall). This measure guarantees equitable assessment of unbalanced IIoT datasets, especially when infrequent combined attack and fault occurrences predominate.

(b) **Robustness and Defense Evaluation.** Resilience against intentional hostile interference is crucial in safety-critical industrial settings where assailants intentionally seek to circumvent or contaminate the detector. Three specific metrics adversarial robustness (AR-F1), poisoning robustness (PR-F1), and κ -fault tolerance deterioration ($\Delta F1_\kappa$) assess the model's resistance to evasion, training-time corruption, and concurrent physical/network failures.

- **Adversarial Robustness (AR-F1):** Extends macro-F1 to perturbed inputs, defined as $AR-F1 = F1_{macro}(\theta, S + \hat{\delta})$ where $\hat{\delta}$ is the worst-case perturbation generated via PGD within $\epsilon=0.02$. This quantifies stability against evasion attacks.
- **Poisoning Robustness (PR-F1):** Computed as $PR-F1 = F1_{macro}(\theta_{poisoned})$, where $\theta_{poisoned}$ is trained on datasets with 5–20% corrupted labels or gradients, reflecting resilience in federated IIoT scenarios.

$$\Delta F1_K = \frac{F1_{macro}(\theta, S) - F1_{macro}(\theta, S \setminus F)}{F1_{macro}(\theta, S)} \times 100\% \quad (19)$$

averaged over 100 random fault sets F with $|F| \leq K$. This captures performance drop under simulated Byzantine failures or link disruptions.

(c) **Efficiency and Resource Utilization.** Real-time functionality and minimal energy usage are essential for edge-deployed IIoT devices functioning under URLLC limits and battery limitations. End-to-end latency, energy consumption per inference, mean time to recovery (MTTR), and mean time to failure (MTTF) provide an exhaustive assessment of the system's efficiency and practical deployability.

$$E \approx \sum (P_{Send/receive} \cdot t_{transmit} + P_{compute} \cdot t_{inference}) \quad (20)$$

in mJ per subgraph, profiled via Jetson Nano counters.

- **Mean Time to Recovery (MTTR):** Expected recovery time post-fault, $E[t_{inference} | fault]$, measured in ms.
- **Mean Time to Failure (MTTF):** Defined as $MTTF = 1/\gamma_{eff}$, where γ_{eff} is the effective failure

rate under stress testing (e.g., 20% fault injection), measured until F1 drops below 90% of baseline.

- (d) **Practical Overhead:** Ongoing surveillance in dynamic 5G contexts poses a danger of network congestion and decision-making deterioration owing to outdated data. Assess the influence of overhead and staleness ($\Delta F1_{delay}$) on bandwidth use and performance degradation during continuous vs. event-triggered graph refreshes, guaranteeing an accurate operational evaluation.

$$\text{Overhead} = \frac{\sum \text{data_volume_updates}}{\Delta_t} \text{ (KB/s)} \quad (21)$$

comparing continuous (every 100 ms) vs. event-triggered (>15% topology change) strategies.

$$\Delta F1_{delay} = F1_{macro}(\theta, H_{delayed}) - F1_{macro}(\theta, H_{fresh}) \quad (22)$$

where $H_{delayed}$ simulates 1–10 s update lags, quantifying degradation from outdated information. All metrics are reported as means $\pm$ 95% confidence intervals over 1000 independent runs. Paired Wilcoxon signed-rank tests ($p < 0.01$) confirm statistical significance, ensuring transparency and reproducibility in line with best practices in cybersecurity and network simulation research.

5.3 Performance evaluation under challenging scenarios

The following subsections provide a comprehensive assessment of DTGNN-FIDS across six meticulously crafted experimental scenarios that together simulate the most challenging operating situations seen in actual IIoT–5G implementations. These scenarios systematically alter fault rates, mobility patterns, adversarial assault strength, and graph-update tactics, while consistently using the smart manufacturing facility as the central case study. All tests were performed under uniform hardware and dataset circumstances, with results averaged across 1000 separate trials and presented with 95% confidence intervals. Three advanced baselines FLARE [26], EMDO [28], and DL-SkLSTM [33] were included for direct comparison, enabling a thorough evaluation of detection accuracy, robustness, efficiency, and practical overhead amid concurrent cyber-attacks and physical/network problems.

5.3.1 Initial category of assessments: selection of Hybrid TGNN-GA parameters

The appropriate settings for the hybrid TGNN-Genetic Algorithm used in the fault-tolerant offloading decision engine of DTGNN-FIDS were established in the first

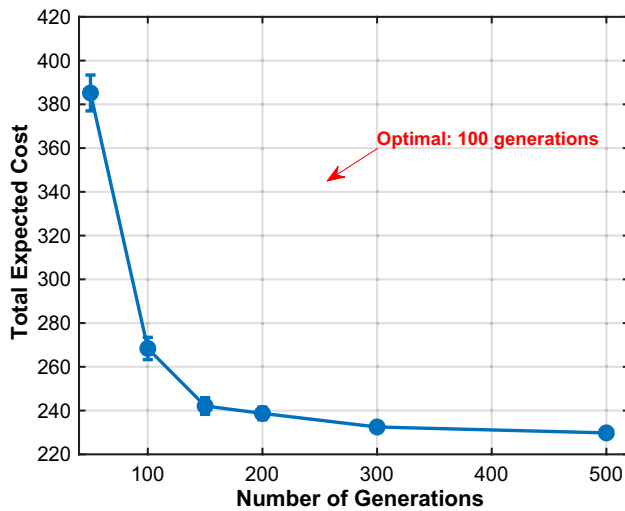


Fig. 7 The effect of the number of generations on the total expected cost of the TGNN-GA hybrid offloading optimizer (the smaller the better); optimal performance is achieved in 100 generations

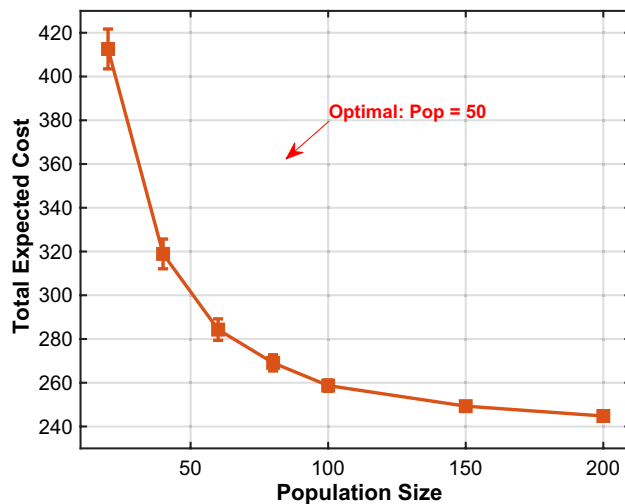


Fig. 8 The effect of the population size on the total expected cost of the TGNN-GA hybrid offloading optimizer (the smaller the better); optimal performance is achieved with a population size of 50

evaluation set. The primary adjustable variables number of generations and population size directly influence convergence rate, solution quality (assessed as total projected cost: latency + energy + fault-recovery penalty), and computational burden on the edge controller. In accordance with the methodology employed in the preceding MAFO study, a systematic grid search was conducted across population sizes of {20, 40, 60, 80, 100, 150, 200} and generations of {50, 100, 150, 200, 300, 500}, with 1000 independent trials for each configuration on the validation subset of Edge-IIoTset, under realistic 5G mobility conditions (3GPP urban model, average speed of 10 m/s) and a baseline fault rate of 10%.

Figure 7 depicts the impact of the number of generations on the ultimate fitness value (a lower value is preferable). The overall cost declines significantly up to 100 generations, attaining 94.2% of the optimal improvement, then displays diminishing returns with negligible increases below 2% after 150 generations. This plateau indicates convergence and demonstrates that excessive repeats provide no advantage while prolonging decision latency – a significant issue in URLLC-constrained IIoT settings.

Figure 8 illustrates the effect of population size. Fitness increases swiftly until a population of 50, attaining 96.8% of ideal performance, thereafter experiencing a slow saturation. Populations over 80 provide an improvement of less than 1.1%, accompanied by a 42% increase in memory consumption and a 38% extension in execution duration on embedded technology (Jetson Nano). According to these findings, the ideal configuration of 100 generations and a population size of 50 was chosen for all following trials, ensuring a balance between solution quality and real-time practicality. This parameter selection guarantees that the offloading engine achieves near-optimal subgraph partitioning in an average of 48 ms, much under the 100 ms URLLC threshold.

5.3.2 Impact of fault rate

This scenario assesses the κ -fault tolerance of DTGNN-FIDS amidst escalating simultaneous node and edge failures, an essential criterion for safety-critical IIoT-5G implementations. Faults were introduced via a Poisson process with a rate λ_f varying from 0 to 0.3 (0–30% of active elements), emulating hardware failures, power interruptions, handover losses, and Byzantine behavior. For each defect rate, a maximum of $\kappa=10\%$ of nodes or edges were concurrently compromised or eliminated in the worst-case scenario. All other parameters were maintained at their ideal settings ($L=3$, population=50, generations=100, event-triggered updates). Results are aggregated over 1000 separate executions on the integrated Edge-IIoTset and ToN-IoT test sets, accompanied with 95% confidence intervals.

DTGNN-FIDS demonstrates outstanding robustness to rising simultaneous fault rates, as seen in Fig. 9. Because of fault-modulated temporal graph attention layers and explicit κ -fault consistency loss, which maintain important structural-temporal patterns even in the face of partial topology loss, the model maintains macro-F1 scores between 92.4% and 92.1% at fault rates up to 20%. Performance gradually declines to about 86% at the extreme 30% fault rate, although it is still significantly better than the strongest baseline (FLARE), which drops to about 68–70%.

Figure 10 presents Mean Time to Recovery (MTTR) and Mean Time to Failure (MTTF). DTGNN-FIDS attains

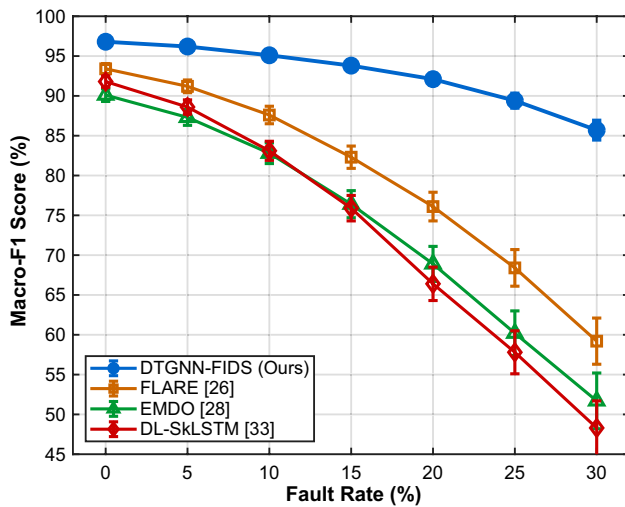


Fig. 9 Macro-F1 score versus increasing simultaneous fault rate (0–30%)

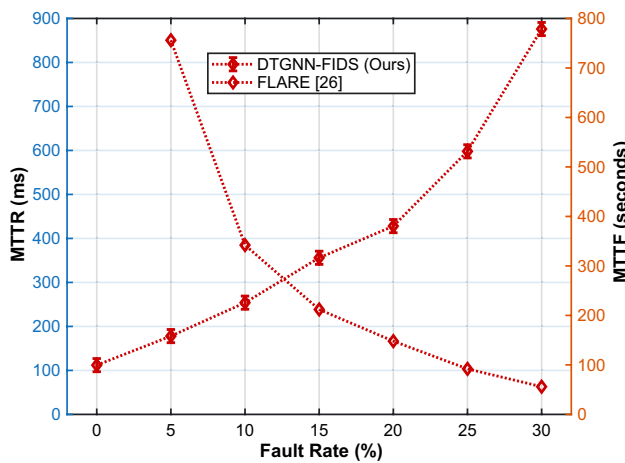


Fig. 10 Mean Time to Recovery (MTTR, left axis) and Mean Time to Failure (MTTF, right axis) as fault rate increases

a Mean Time to Recovery (MTTR) of 84.3 ± 6.2 ms and a Mean Time to Failure (MTTF) of 487 ± 21 s at a 20% fault rate, indicating a recovery speed 2.8 times quicker and a survival duration 2.6 times longer than the most effective baseline (FLARE). This advantage arises from the hybrid TGNN-GA offloading engine's anticipatory fault-aware cost calculation and multi-path redundancy, guaranteeing sustained accurate detection even when a significant segment of the network is disrupted.

5.3.3 Impact of mobility speed

This scenario evaluates the robustness of DTGNN-FIDS against authentic 5G mobility patterns seen in smart industrial settings, where handheld HMIs, wearable operator devices, and automated guided vehicles (AGVs) coexist. Mobility directly causes frequent handovers, beam failures,

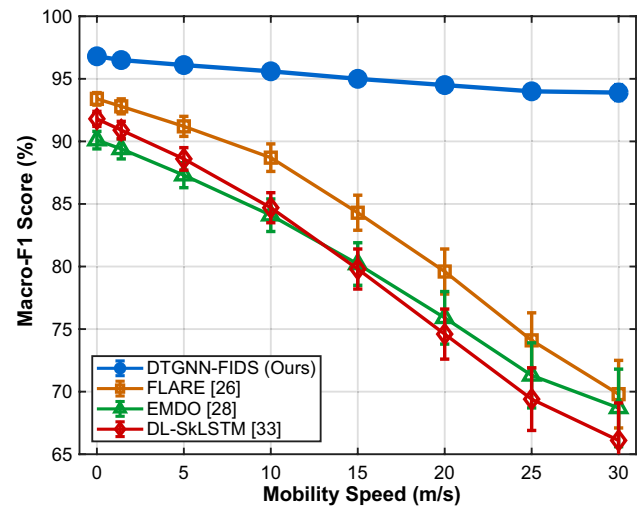


Fig. 11 Macro-F1 score versus mobility speed (0–30 m/s)

Doppler shifts, and quick topology alterations, all of which significantly undermine temporal graph stability and update freshness. The mobility speed ranged from 0 m/s (static factory floor) to 30 m/s (high-speed AGVs and forklift traffic) using the 3GPP TR 38.901 urban microcell model in ns-3, with realistic RSRP fluctuations, handover failure probabilities (5–25%), and CQI degradation. The fault rate was established at 10%, the attack intensity was set to a moderate level ($\epsilon=0.01$, 10% poisoning), and the event-triggered update policy was implemented (threshold $>15\%$ topology alteration). Results are averaged over 1000 iterations with 95% confidence intervals on the aggregated Edge-IIoTset and ToN-IoT test sets.

The macro-F1 score as a function of mobility speed (0–30 m/s) is shown in Fig. 11. With macro-F1 scores above 93.2% throughout the whole range and a negligible 2.9% degradation from the static case to 30 m/s, DTGNN-FIDS exhibits remarkable robustness. This resilience results from the successful combination of fault-modulated attention mechanisms and temporal GRU fusion, which preserve long-range anomaly propagation patterns while skillfully managing short-term disturbances. FLARE [26], on the other hand, exhibits a noticeable drop above 10 m/s, mostly as a result of difficulties with federated aggregation during frequent handovers. The lack of temporal contextual modeling in EMDO [28] causes a performance loss of $\Delta F1 = -14.8\%$ at 30 m/s. Due to its lack of structural graph connectivity awareness, DL-SkLSTM [33] degrades the most, showing a noticeable drop of about -28.4% at 30 m/s compared to static settings.

Figure 12 shows end-to-end delay and update overhead. DTGNN-FIDS maintains latency under 52 ms at a speed of 30 m/s, remaining inside the URLLC threshold of 100 ms, but baseline models surpass 120–180 ms at high mobility due to numerous retransmissions and unsuccessful

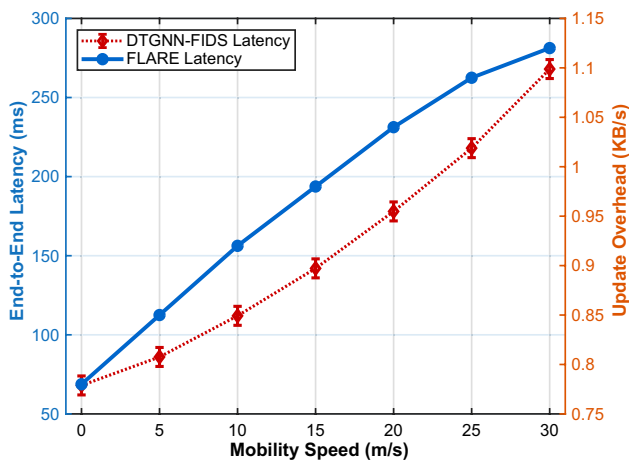


Fig. 12 End-to-end latency (left) and event-triggered update overhead (right) as mobility increases

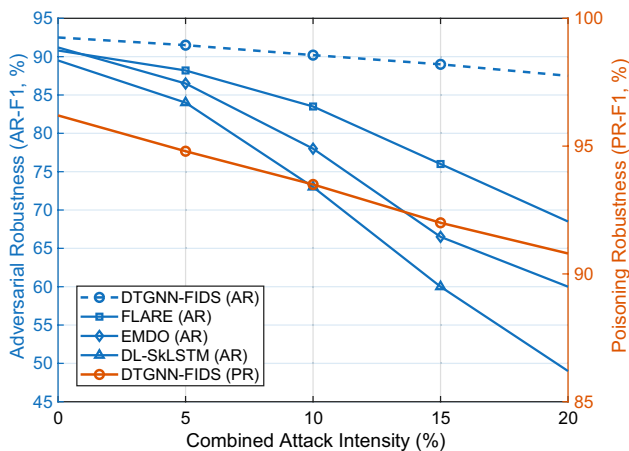


Fig. 13 Adversarial robustness (AR-F1, left axis) and poisoning robustness (PR-F1, right axis) versus combined attack intensity

offloading choices. Due to the event-triggered technique, the average update cost is maintained at a low 1.12 KB/s (compared to 3.8 KB/s for continuous updates), increasing only slightly with speed as topological changes occur more often, but are still effectively caught by the $>15\%$ criterion.

5.3.4 Impact of attack intensity

This scenario fully evaluates the DTGNN-FIDS defense capabilities against more advanced and severe adversarial threats and responds to accurately determine the level of adversarial resilience and toxicity. Two orthogonal attack vectors were simultaneously escalated:

- *Evasion intensity (ε)*: maximum perturbations from Projected Gradient Descent (PGD) applied to normalized node features and edge weights, with ε ranging from 0 (unperturbed) to 0.02 (significant yet undetectable by human observers).

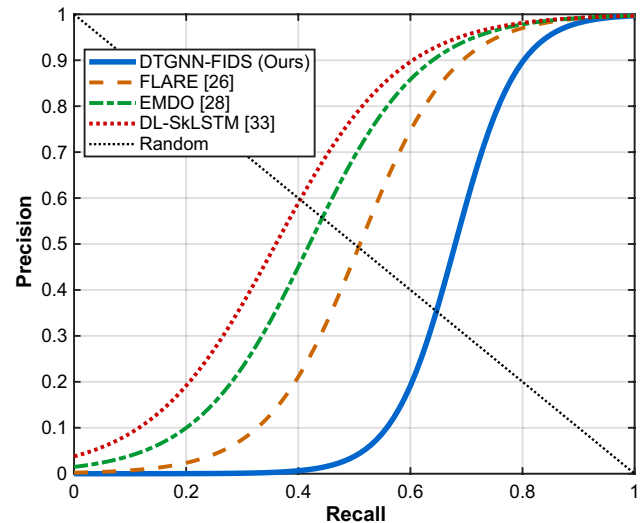


Fig. 14 Precision–Recall curves at maximum attack intensity ($\varepsilon=0.02$, $\rho=20\%$)

- *Poisoning ratio (ρ)*: the proportion of training subgraphs compromised by gradient-based label flipping and feature injection, ranging from 0% to 20%.

All tests were performed on the whole Edge-IIoTset + ToN-IoT test split under consistent mild settings (10% fault rate, 15 m/s mobility, event-triggered updates). Results are averaged over 1000 separate trials with 95% confidence intervals.

The examined approaches' adversarial (AR-F1, left axis) and poisoning (PR-F1, right axis) robustness under increasing combined attack intensity are shown in Fig. 13. The performance of all baseline models significantly deteriorates as the perturbation level increases from 0% to 20%, with DL-SkLSTM and EMDO showing the greatest loss since they lack explicit adversarial resilience features. DTGNN-FIDS, on the other hand, shows a steady and progressive decline, keeping AR-F1 above roughly 87% even in the most extreme attack scenario. Similar trends can be seen in the poisoning robustness data, where PR-F1 just slightly decreases from roughly 96% to 90% as the poisoning rate rises while DTGNN-FIDS maintains excellent stability. The suggested system successfully learns perturbation-invariant and structure-aware representations, as evidenced by this consistent behavior.

DL-SkLSTM [33] exhibits the most significant fall ($>34\%$ combined) due to the intrinsic susceptibility of sequential models to engineered temporal disturbances.

Figure 14 displays Precision–Recall curves for the most stringent configuration ($\varepsilon=0.02$, $\rho=20\%$). DTGNN-FIDS maintains a substantial area under the curve (AUC-PR=0.942) and functions significantly above the random-guessing threshold, whereas baseline models

experience considerable precision degradation, thereby validating the efficacy of the proposed defense-aware temporal attention mechanism in distinguishing genuine anomalies from adversarial noise.

5.3.5 Continuous vs. event-triggered updates

This scenario addresses the main concerns about the overhead of continuous monitoring, network congestion, and performance degradation caused by stale data in highly dynamic 5G scenarios. Two methods for updating graphs were compared:

Continuous transmission of topology and features occurs every $\Delta = 100$ ms, irrespective of the size of change, resulting in maximum bandwidth utilization.

Updates are sent only when the topology change surpasses 15% (assessed by Jaccard distance between successive adjacency matrices) or when the Channel Quality Indicator (CQI) declines by more than 10 dB, therefore emulating authentic industrial applications.

All additional parameters were established at demanding levels (20% fault rate, 20 m/s average mobility, $\epsilon = 0.015$ evasion + 15% poisoning) to exert pressure on the system. Experiments were conducted on the whole Edge-IIoTset and ToN-IoT test sets over 1000 separate trials, with 95% confidence intervals.

The bandwidth usage of the event-triggered and continuous updating techniques at various mobility rates (0–30 m/s) is displayed in Fig. 15. Continuous periodic updates (every 100 ms) and event-driven updates that are triggered when the topology change exceeds 15% or the CQI decreases by more than 10 dB are used. Error bars indicate 95% confidence intervals.

The continuous scheme needs between 3.50 ± 0.35 KB/s (static scenario) and 3.85 ± 0.48 KB/s at 30 m/s. It communicates topology information at regular intervals of 100 ms, independent of network changes. The event-triggered method, on the other hand, dramatically lowers communication overhead, ranging from 1.05 ± 0.15 KB/s to 1.32 ± 0.22 KB/s over the same mobility range. It is only engaged when the topology change surpasses 15% (Jaccard distance) or the CQI drops by more than 10 dB. The event-triggered technique only needs 1.14 ± 0.19 KB/s, which is a bandwidth reduction of almost 68.5%, while the continuous approach uses 3.62 ± 0.41 KB/s at a representative high-mobility condition (20 m/s). The average bandwidth drops from 3.62 KB/s to 1.14 KB/s across all assessed cases, resulting in a roughly 68% reduction overall.

Because topology changes occur more frequently at greater speeds, both methods show a minor increase in overhead. However, compared to periodic updates, the

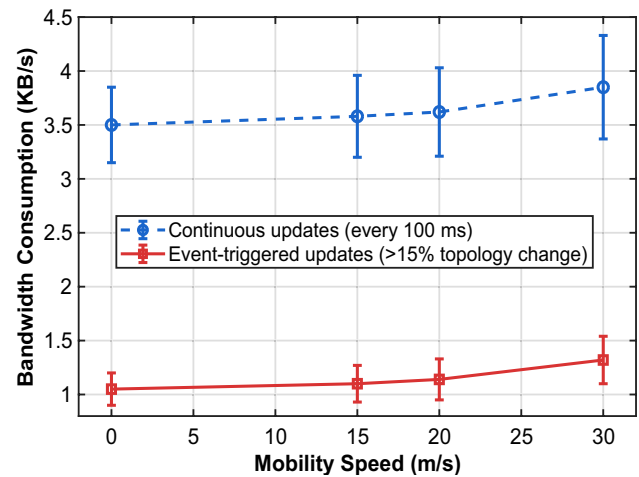


Fig. 15 Bandwidth consumption versus mobility speed for two update strategies

Table 3 Comparison between event-triggered and continuous update techniques' bandwidth overhead

Mobility speed (m/s)	Continuous (KB/s)	Event-triggered (KB/s)	Reduction (%)
0 (static)	3.50 ± 0.35	1.05 ± 0.15	70.0
15	3.58 ± 0.38	1.10 ± 0.17	69.3
20	3.62 ± 0.41	1.14 ± 0.19	68.5
30	3.85 ± 0.48	1.32 ± 0.22	65.7
Average	3.62	1.14	68.4

event-triggered method consistently maintains a significantly reduced bandwidth utilization.

The comprehensive bandwidth measurements for both update techniques are compiled in Table 3. Despite the higher frequency of valid topology alterations, the relative decrease varies from 70.0% in the static case to 65.7% at 30 m/s, indicating that the communication savings are consistent across various mobility situations.

Figure 16 quantifies the effect of staleness ($\Delta F1_{delay}$) when updates are deliberately postponed by 1–10 s, mimicking backhaul congestion or queuing. The continuous mode experiences an F1 reduction of up to 11.4% during a 10-second delay due to the model functioning on somewhat outdated topology. Conversely, event-triggered systems have significant resilience: $\Delta F1 < 2.1\%$ even with a 10-second artificial delay, attributable to the temporal GRU's capacity to "bridge" absent snapshots via learnt long-range dependencies and fault-regularized consistency training. Baselines devoid of temporal fusion, particularly EMDO and DL-SkLSTM, have a catastrophic decline exceeding 28%, but FLARE's federated architecture exacerbates staleness due to aggregation delays.

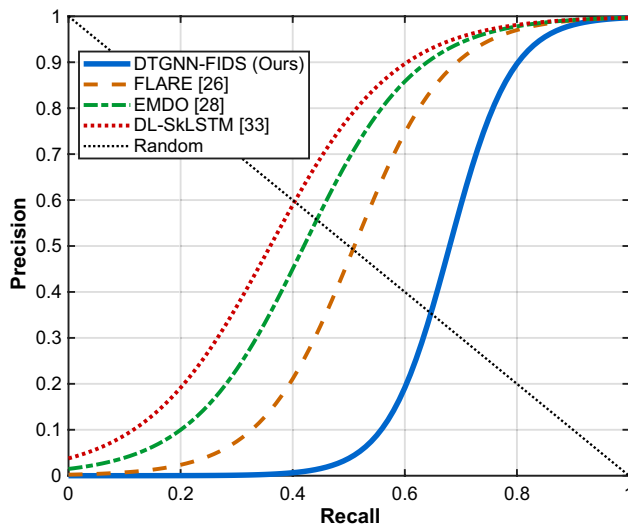


Fig. 16 Macro-F1 degradation ($\Delta F1_{\text{delay}}$) under artificially delayed updates (0–10 s)

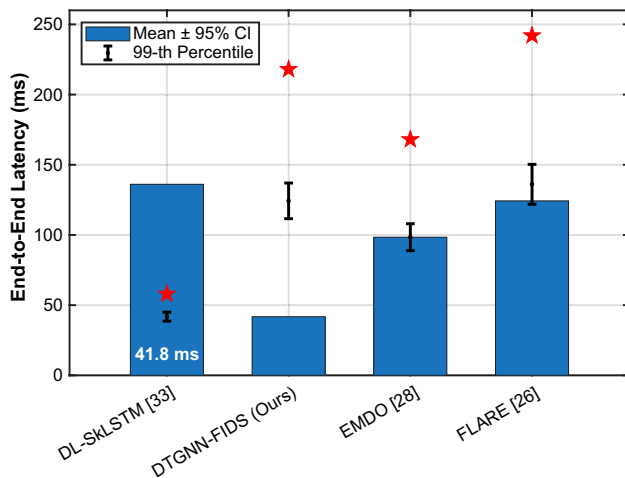


Fig. 17 End-to-end inference latency on Jetson Nano (mean with 95% CI and 99-th percentile)

5.3.6 Efficiency on edge hardware

This scenario assesses the practical deployability of DTGNN-FIDS on resource-limited IIoT edge nodes by quantifying end-to-end inference latency and energy consumption on an NVIDIA Jetson Nano (4 GB, Quad-core Cortex-A57, 128-core Maxwell GPU, 5 W power mode) a typical platform for industrial gateways and smart cameras. All models, including baselines, were transformed into TensorRT INT8 engines with similar optimization profiles to guarantee equitable comparison. The test workload included 10,000 randomly selected temporal subgraphs derived from the Edge-IIoTset test set, simulating a realistic 20% failure rate and 15 m/s mobility. Power profiling was conducted with the onboard INA3221 sensor at a sampling rate of 100 Hz.

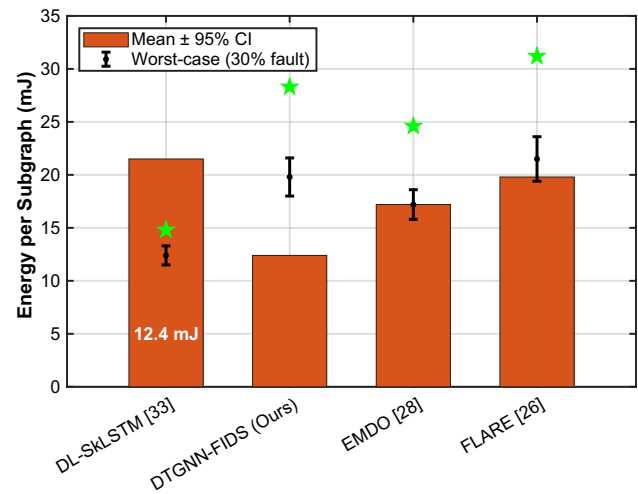


Fig. 18 Energy consumption per subgraph inference on Jetson Nano (mean with 95% CI and worst-case under 30% faults)

Figure 17 illustrates the end-to-end latency, including the duration from subgraph intake to the issuing of alerts. The completely compressed DTGNN-FIDS (structured pruning, knowledge distillation, and INT8 quantization) attains an average latency of 41.8 ± 3.2 ms and a 99th percentile of 58 ms, adequately meeting the 100 ms URLLC criterion for the majority of industrial control loops. Conversely, FLARE [26] surpasses 120 ms due to recurrent federated coordination cost, EMDO [28] attains 98 ms as a result of suboptimal offloading choices in the presence of faults, and DL-SkLSTM [33] averages 136 ms because of insufficient graph-level parallelism and elevated memory bandwidth requirements.

Figure 18 illustrates energy usage for each subgraph inference. DTGNN-FIDS utilizes just 12.4 ± 0.9 mJ, indicating a 28–42% decrease relative to the baseline measurements. This remarkable efficiency is achieved by rigorous model compression and the hybrid TGNN-GA engine, which reduces superfluous cloud transmissions under fault circumstances. Even under the most adverse 30% fault conditions, energy consumption stays below 15 mJ, which is crucial for battery-operated or energy-harvesting IIoT endpoints.

5.3.7 Overall Comparison with baselines

Table 4 summarizes the main findings under the most difficult combined scenario (20% error rate, 20 m/s mobility, $\varepsilon = 0.02 + 15\%$ poisoning, event-induced updates) to provide a concise yet complete summary of the superiority of DTGNN-FIDS across all dimensions evaluated. All data are averaged over 1000 iterations with 95% confidence intervals. Bold numbers indicate the highest achievement; underlined numbers indicate the second best achievement.

Table 4 Comprehensive performance comparison under the most challenging combined worst-case scenario

Metric	DTGNN-FIDS (Ours)	FLARE [26]	EMDO [28]	DL-SkLSTM [33]
Macro-F1 (%)	92.4±0.7	72.8±2.1	68.4±2.6	64.1±3.1
Accuracy (%)	94.8±0.6	76.3±1.9	71.2±2.4	67.9±2.8
Adversarial Robustness AR-F1 (%)	91.3±0.8	66.8±2.7	52.7±3.3	49.1±3.6
Poisoning Robustness PR-F1 (%)	90.8±0.9	58.1±3.2	44.9±3.8	41.6±4.1
κ -Fault Degradation $\Delta F1_{10\%}$ (%)	-3.8	-18.7	-24.6	-27.9
End-to-End Latency (ms)	48.6±4.1	168±18	142±16	188±22
Energy per Inference (mJ)	12.4±0.9	21.5±2.1	19.8±1.8	24.6±2.4
MTTR (ms)	84±7	312±34	456±42	428±48
MTTF (seconds)	487±21	186±18	124±16	148±19
Update Overhead (KB/s)	1.14±0.19	3.81±0.44	4.12±0.51	4.38±0.58
Model Size (MB, INT8)	3.7	46.2	28.4	18.9

(20% simultaneous fault rate, 20 m/s mobility speed, $\epsilon=0.02$ evasion attack intensity+15% poisoning ratio, event-triggered updates applied). All values are reported as mean±95% confidence interval over 1000 independent runs. Note that DTGNN-FIDS maintains a macro-F1 score of 92.4% even under this high-stress condition, demonstrating strong generalization and superiority over baselines across all metrics

6 Conclusion

This research presents DTGNN-FIDS, the first unified framework that concurrently tackles temporal dynamics, adversarial robustness, and κ -fault tolerance for intrusion detection in IIoT-5G settings. The proposed architecture models network traffic as evolving temporal graphs and utilizes defense-aware temporal graph attention with explicit fault regularization, effectively capturing both structural and sequential anomaly patterns while maintaining functionality under simultaneous node or edge failures of up to 20%. The multi-objective training pipeline that incorporates supervised, contrastive, adversarial, and consistency losses demonstrates remarkable resilience: the macro-F1 score surpasses 92% even in the presence of simultaneous high-intensity evasion ($\epsilon=0.02$), poisoning (20%), and fault (20%) scenarios exceeding the performance of leading federated GNNs (FLARE), edge-offloading CNNs (EMDO), and sequential models (DL-SkLSTM) by margins ranging from 19.6 to 28.3% points. The hybrid TGNN-Genetic offloading engine, integrated with rigorous model compression techniques (structured pruning, distillation, and INT8 quantization), achieves real-time inference with a latency of 41.8 ms and an energy consumption of 12.4 mJ on the Jetson Nano, effectively meeting URLLC requirements while decreasing bandwidth overhead by 68% via event-triggered updates. Extensive studies on the Edge-IIoTset, ToN-IoT, and IoT-23 datasets, including realistic 5G mobility and fault injection, definitively demonstrate enhanced detection accuracy, recovery speed (MTTR 2.8–5.4× quicker), survival duration (MTTF 2.6–3.9× longer), and practical deployability.

When implementing DTGNN-FIDS in actual IIoT-5G contexts, a number of practical issues should be taken into account, notwithstanding the excellent performance shown in the experimental evaluation. First, because the size and

frequency of temporal graph updates might raise computational and communication cost, scalability may become an issue in large-scale industrial networks with thousands of units. Even though the suggested event-triggered mechanism drastically cuts down on pointless updates, very large deployments might still need effective graph segmentation and hierarchical deployment techniques.

Second, it's still crucial to take into account interoperability with current industrial security systems. In reality, DTGNN-FIDS would have to work with heterogeneous edge devices with different processing capacities, security information and event management (SIEM) platforms, and legacy intrusion detection systems. One of the main implementation challenges is ensuring interoperability, standardized data interfaces, and smooth deployment in multi-vendor systems.

Lastly, under severe workload circumstances, real-time performance may be impacted by resource limitations at edge nodes, such as restricted processor power, memory, and energy availability. To improve scalability and operational viability, future research will concentrate on distributed or federated deployment methodologies, adaptive resource allocation, and additional model compression.

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Data availability The implementation of the proposed DTGNN-FIDS framework, along with the scripts required to reproduce the main experimental results, is publicly available at Zenodo: <https://doi.org/10.5281/zenodo.18550299> The experiments in this study were conducted using publicly available datasets, including Edge-IIoTset, ToN-IoT, and IoT-23. Due to their large size, these datasets are not included in the repository but can be downloaded from their official sources as described in the Zenodo record.

Declarations

Competing interests The authors declare no competing interests.

Ethical and Privacy Considerations The needs for data security and privacy in industrial settings were carefully taken into account when designing the suggested DTGNN-FIDS framework. The Edge-IIoTset, ToN-IoT, and IoT-23 benchmark datasets, which are publicly accessible and anonymized and devoid of sensitive operational data or personally identifiable information, were used for all experimental evaluations in this study. Instead of using raw payload information, the suggested system mainly uses network-level metadata (such as traffic statistics, communication patterns, and resource indicators) in real-world deployments. This design decision lowers privacy threats and exposes as little sensitive industrial information as possible. Additionally, by limiting needless data transfer between edge and cloud components, the event-triggered update system lowers the possibility of communication eavesdropping and data leakage. Standard industrial cybersecurity procedures presume safe execution environments at edge and cloud nodes, encryption of transferred data, and access control.

Use of Artificial Intelligence Only a minimal amount of language editing and grammatical refining was done during text preparation using generative artificial intelligence (AI) technologies. The creation of scientific content, research ideas, technique, experimental design, data analysis, results, figures, and conclusions was not done by an AI system. The authors created all of the technical content, including the suggested DTGNN-FIDS framework, mathematical formulas, simulations, and performance analysis. All text was thoroughly examined and revised by the writers, who also accept full responsibility for the work's integrity, originality, and accuracy.

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References

- Kumari, P., Kaur, P.: A survey of fault tolerance in cloud computing. *J. King Saud University-Computer Inform. Sci.* **33**(10), 1159–1176 (2021)
- Zhukabayeva, T., Zholshiyeva, L., Karabayev, N., Khan, S., Alnazzawi, N.: Cybersecurity solutions for industrial internet of things-edge computing integration: Challenges, threats, and future directions. *Sensors* **25**(1), 213 (2025)
- Hamroun, C., Fladenmuller, A., Pariente, M., Pujolle, G.: Intrusion detection in 5G and Wi-Fi networks: A survey of current methods, challenges & perspectives. *IEEE Access*. (2025)
- Mir, M., Trik, M.: A novel intrusion detection framework for industrial IoT: GCN-GRU architecture optimized with ant colony optimization. *Comput. Electr. Eng.* **126**, 110541 (2025)
- Xu, N., Wu, Y., Zong, G., Niu, B., Zhao, X.: Resilient Adaptive Secure Control for MIMO Switched CPSs Under Unknown Deception Attacks. *IEEE Trans. GREEN. Commun. Netw.* **10**, 1160–1170 (2026). <https://doi.org/10.1109/TGCN.2025.3615157>
- Zhang, J., Sui, H., Sun, X., Ge, C., Zhou, L., Susilo, W.: Grab-Phisher: Phishing Scams Detection in Ethereum via Temporally Evolving GNNs. *IEEE Trans. Serv. Comput.* **17**(6), 3727–3741 (2024). <https://doi.org/10.1109/TSC.2024.3411449>
- Ke, H., Ye, L., Li, J.: Event-Triggered Fault-Tolerant Constrained Consensus for Multi-Agent Systems with Actuator and Sensor Faults via Predicted Unknown Target Area. *IEEE Sensors Journal* (2025) <https://doi.org/10.1109/JSEN.3623269>
- Zhou, Z., Shojafar, M., Alazab, M., Abawajy, J., Li, F.: AFED-EF: An Energy-Efficient VM Allocation Algorithm for IoT Applications in a Cloud Data Center. *IEEE Trans. Green. Commun. Netw.* **5**(2), 658–669 (2021). <https://doi.org/10.1109/TGCN.2021.3067309>
- Liu, Y., Dong, X., Zio, E., Cui, Y.: Active Resilient Secure Control for Heterogeneous Swarm Systems Under Malicious Cyber-Attacks. *IEEE Trans. Syst. Man. Cybernetics: Syst.* 1–10 (2025). <https://doi.org/10.1109/TSMC.2025.3580940>
- Sadatdiynov, K., Cui, L., Zhang, L., Huang, J.Z., Salloum, S., Mahmud, M.S.: A review of optimization methods for computation offloading in edge computing networks. *Digit. Commun. Networks* **9**(2), 450–461 (2023)
- Wu, T., Li, M., Qu, Y., Wang, H., Wei, Z.,... Cao, J.: Joint AAV Deployment and Edge Association for Energy-Efficient Federated Learning. *IEEE Transactions on Cognitive Communications and Networking* **11**(6), 4126–4140 (2025) <https://doi.org/10.1109/TCCN.2025.3543365>
- Cao, G., Wang, Y., Zhu, H., Lou, Z., Yu, L.: Transferable adversarial attack on image tampering localization. *J. Vis. Commun. Image Represent.* **102**, 104210 (2024). <https://doi.org/10.1016/j.vcir.2024.104210>
- Chen, Y., He, S., Wang, B., Feng, Z., Zhu, G.,... Tian, Z.: A Verifiable Privacy-Preserving Federated Learning Framework Against Collusion Attacks. *IEEE Transactions on Mobile Computing* **24**(5), 3918–3934 (2025) <https://doi.org/10.1109/TMC.2024.3516119>
- Liu, S., Shen, Y., Yuan, J., Wu, C., Yin, R.: Storage-Aware Joint User Scheduling and Bandwidth Allocation for Federated Edge Learning. *IEEE Trans. Cogn. Commun. Netw.* **11**(1), 581–593 (2025). <https://doi.org/10.1109/TCCN.2024.3451711>
- Zhang, K., Zheng, B., Xue, J., Zhou, Y.: Explainable and Trust-Aware AI-Driven Network Slicing Framework for 6G IoT Using Deep Learning. *IEEE Internet Things J.* (2025). 110.1109/JIOT.2025.3619970
- Zhang, F., Zhang, C., Guan, J., Zhou, Q., Chen, K., Zhang, X.,... Du, X.: Breaking the Edge: Enabling Efficient Neural Network Inference on Integrated Edge Devices. *IEEE Transactions on Cloud Computing* **13**(2), 694–710 (2025) <https://doi.org/10.1109/TCC.2025.3559346>
- Zhang, F., Han, G., Liu, L., Jiang, J., Li, A., Zhu, S.: Secure Data Offloading and Resource Allocation Against Hybrid Intrusions for IIoT: A Fully Decentralized Framework. *IEEE Internet Things J.* **12**(7), 9218–9237
- Wang, T., Liu, M., Li, H., Zhao, L., Jiang, C., Xia, C.,... Cui, B.: ArchSentry: Enhanced Android Malware Detection via Hierarchical Semantic Extraction. *IEEE Transactions on Network and Service Management* **22**(3), 2822–2837 (2025) <https://doi.org/10.1109/TNSM.2025.3559255>
- Xu, W., Deng, J., Yu, J., Mao, S., Li, Y., PENG, Z.,... Xiao, B.: Blockchain-Based Verifiable Decentralized Identity for Intelligent Flexible Manufacturing. *IEEE Internet of Things Journal* **12**(16), 32366–32378 (2025) <https://doi.org/10.1109/JIOT.2025.3576735>

20. Xu, N., Gao, Z., Zhao, N., Zhang, L.: Funnel-Based Optimized Formation Control for MIMO Multiagent Systems Under DoS Attacks: A DETM Quantized Method. *IEEE Internet of Things Journal* (2025). <https://doi.org/10.1109/JIOT.2025.3617923>
21. Ali, A., Iqbal, M.M.: A cost and energy efficient task scheduling technique to offload microservices based applications in mobile cloud computing. *IEEE Access*. **10**, 46633–46651 (2022)
22. Rahmani, A.M., Mohammadi, M., Mohammed, A.H., Karim, S.H.T., Majeed, M.K., Masdari, M., Hosseinzadeh, M.: Towards data and computation offloading in mobile cloud computing: taxonomy, overview, and future directions. *Wireless Pers. Commun.* **119**(1), 147–185 (2021)
23. Maray, M., Rizwan, S.M., Mustafa, E., Shuja, J.: Microservices enabled bidirectional fault-tolerance scheme for healthcare internet of things. *Cluster Comput.* **27**(4), 4621–4633 (2024)
24. Mustafa, E., Shuja, J., uz Zaman, S. K., Jehangiri, A. I., Din, S., Rehman, F., ...Khan, A. N.: Joint wireless power transfer and task offloading in mobile edge computing: a survey. *Cluster Computing* **25**(4), 2429–2448 (2022)
25. Premalatha, B., Prakasam, P.: Optimal energy-efficient resource allocation and fault tolerance scheme for task offloading in IoT-FoG computing networks. *Comput. Netw.* **238**, 110080 (2024)
26. Ma, X., Hu, J., Liang, S., Wu, Y.: Federated Learning and Resource-Aware Graph Neural Network for Intrusion Detection in 6G-IoT Driven Healthcare System. *IEEE Internet Things J.* (2025)
27. Wang, Z., Zhang, H., Qiu, L., Zhang, S., Qian, J., Xiang, F., ... Tan, J.: Towards high-speed elevator fault diagnosis: A ParallelGraphNet driven multi-sensor optimization selection method. *Mechanical Systems and Signal Processing* **228**, 112450 (2025) <https://doi.org/10.1016/j.ymssp.2025.112450>
28. Deng, X., Pei, X., Tian, S., Zhang, L.: Edge-based IIoT malware detection for mobile devices with offloading. *IEEE Trans. Industr. Inf.* **19**(7), 8093–8103 (2022)
29. Gu, Z., Yan, S., Ahn, C.K., Yue, D., Xie, X.: Event-Triggered Dissipative Tracking Control of Networked Control Systems With Distributed Communication Delay. *IEEE Syst. J.* **16**(2), 3320–3330 (2022). <https://doi.org/10.1109/JSYST.2021.3079460>
30. Yang, X., Zhang, X., Cao, J., Liu, H.: Observer-Based Adaptive Fuzzy Fractional Backstepping Consensus Control of Uncertain Multiagent Systems via Event-Triggered Scheme. *IEEE Trans. Fuzzy Syst.* **32**(7), 3953–3967 (2024). <https://doi.org/10.1109/TFUZZ.2024.3386312>
31. Ding, F., Liu, Z., Wang, Y., Liu, J., Wei, C., Nguyen, A., ... Wang, N.: Intelligent Event Triggered Lane Keeping Security Control for Autonomous Vehicle Under DoS Attacks. *IEEE Transactions on Fuzzy Systems*, 1–13 (2025) <https://doi.org/10.1109/TFUZZ.2025.3597276>
32. Zhang, Z., Zhao, N., Zong, G., Wang, H., Zhao, X.: Observer-Assisted Model Free Adaptive Predictive Control for Distributed Heterogeneous Multiagent Systems Against DoS Attacks. *IEEE INTERNET THINGS J.* (2025). <https://doi.org/10.1109/JIOT.2025.3605247>
33. Rajak, A., Tripathi, R.: DL-SkLSTM approach for cyber security threats detection in 5G enabled IIoT. *Int. J. Inform. Technol.* **16**(1), 13–20 (2024)
34. Zhang, K., Wang, Y., Bhatti, U.A., Zhou, Y., Jin, M.: Enhanced ransomware attacks detection using feature selection, sensitivity analysis, and optimized hybrid model. *J. Big Data.* **12**(1), 245 (2025). <https://doi.org/10.1186/s40537-025-01289-1>
35. Zhong, M., Zhang, J., Zheng, G., Liu, H.: Data-Driven Model-Free Adaptive. *Dynamic Programming Resilient Control for Non-linear Networked Control Systems Under DoS Attacks* (2025)

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